Planned Sprint	Actual Sprint	US ID	User Story Description	MOSCOW	Dependency	Assignee	Status
S1	S1	US-001	Project Kickoff and Repository Setup – Initialize GitHub repository, define folder structure, assign team roles and responsibilities.	MUST HAVE	No Dependency	Leena Gavande	3- Completed
S1	S1	US-001	Workflow Creation and Architecture Design – Design system architecture diagram showing data flow from API to ML to FastAPI to React frontend.	MUST HAVE	Lucid Chart / Draw. io	Gowthama R.	3- Completed
S2	S2	US-001	Agile Document Preparation – Prepare Product Backlog, Sprint Backlog, Stand-up, and Retrospection templates for Agile tracking.	MUST HAVE	No Dependency	Gowthama R.	3- Completed
S2	S2	US-001	Financial API Integration – Research and integrate Yahoo Finance / Alpha Vantage API to fetch historical stock and ETF price and volume data.	MUST HAVE	Yahoo Finance / Alpha Vantage API	Leena Gavande	3- Completed
S2	S2	US-001	Data Preprocessing Pipeline – Clean raw market data using Pandas: handle missing values, remove outliers, and engineer technical indicators.	MUST HAVE	Pandas, NumPy	Leena Gavande	3- Completed
S3	S3	US-001	ML Model Development – Train Random Forest and XGBoost models on historical data to classify Buy, Sell, and Hold signals with evaluation metrics.	MUST HAVE	Scikit-learn, XGBoost	Leena Gavande	3- Completed
S4	S4	US-001	FastAPI Backend Development – Design and build RESTful API endpoints for data retrieval, ML model inference, and signal delivery.	MUST HAVE	FastAPI, Uvicorn	Gowthama R.	3- Completed
S5	S5	US-001	ML and Backend Integration – Integrate preprocessing pipeline and ML model into FastAPI for real-time Buy/Sell/Hold signal prediction via /predict.	MUST HAVE	FastAPI, Scikit-learn, XGBoost	Gowthama R.	3- Completed
S6	S6	US-001	React Frontend Dashboard – Build interactive React dashboard with stock charts, signal display tables, and symbol/date-range filter controls.	MUST HAVE	React.js, Recharts, Axios	K. Ashok	2- In Progress
S7	S7	US-001	Frontend–Backend Integration – Connect React frontend to FastAPI backend using Axios API calls; handle CORS, loading states, and error responses.	MUST HAVE	React.js, FastAPI, CORS Middleware	K. Ashok	2- In Progress
S8	S8	US-001	Deployment and CI/CD – Deploy FastAPI backend on Render and React frontend on Vercel; configure GitHub Actions for automated CI/CD pipelines.	MUST HAVE	Render, Vercel, GitHub Actions	Karishma K.	2- In Progress

NOTES: Task sizing should be between 0.5 to 12 hours

US ID	Task ID	Task Description	Task Start Date	Task Completion Date	Team Member	Activity	Status	Original Estimate Effort (In Hours)	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9	Day 10	Day 11	Day 12	Day 13	Day 14
								102	16	7	10	4	8	5	6	4	6	3	5	8	6	14
SPRINT 1 BACKLOG																						
US-001	T-01	Project Kickoff – Initialize GitHub repo, define folder structure and assign team roles	2/Feb/26	4/Feb/26	Gowthama R.	Others	3- Completed	3	1	1	1	0	0	0	0	0	0	0	0	0	0	0
US-001	T-02	Workflow Creation – Design system architecture and data flow diagram using Draw.io	5/Feb/26	7/Feb/26	Gowthama R.	Design	3- Completed	4	3	1	0	0	0	0	0	0	0	0	0	0	0	0
US-001	T-03	Agile Document Preparation – Set up Product Backlog, Sprint Backlog, Stand-up, and Retrospection sheets	9/Feb/26	9/Feb/26	Gowthama R.	Others	3- Completed	2	2	0	0	0	0	0	0	0	0	0	0	0	0	0
US-001	T-04	Financial API Research – Evaluate Yahoo Finance and Alpha Vantage APIs for data availability and rate limits	10/Feb/26	11/Feb/26	Leena Gavande	Others	3- Completed	3	0	0	2	1	0	0	0	0	0	0	0	0	0	0
US-001	T-05	API Data Fetching – Write Python scripts to pull historical OHLCV data for stocks and ETFs	12/Feb/26	13/Feb/26	Leena Gavande	Coding	3- Completed	5	0	0	0	0	3	2	0	0	0	0	0	0	0	0
SPRINT 2 BACKLOG																						
US-001	T-06	Data Preprocessing – Clean missing values, normalize data, and apply forward-fill for gaps using Pandas	2026-02-16 0:00:00	18-Feb-26	Leena Gavande	Coding	3- Completed	5	0	0	0	0	0	0	3	2	0	0	0	0	0	0
US-001	T-07	Feature Engineering – Compute technical indicators: RSI, MACD, Bollinger Bands, and Moving Averages	2026-02-19 0:00:00	21-Feb-26	Leena Gavande	Coding	3- Completed	6	0	0	0	0	0	0	0	0	3	2	1	0	0	0
US-001	T-08	Random Forest Model Training – Train and evaluate Random Forest classifier for Buy/Sell/Hold signal generation	2026-02-23 0:00:00	25-Feb-26	Leena Gavande	Build	3- Completed	8	0	0	0	0	0	0	0	0	0	0	0	4	4	0
US-001	T-09	XGBoost Model Training – Train XGBoost model, tune hyperparameters, and compare accuracy with Random Forest	2026-02-26 0:00:00	27-Feb-26	Leena Gavande	Build	3- Completed	8	0	0	0	0	0	0	0	0	0	0	0	0	0	8
SPRINT 3 BACKLOG																						
US-001	T-10	FastAPI Project Setup – Create FastAPI project structure with routers, schemas, and environment config	2026-03-02 0:00:00	3-Mar-26	Gowthama R.	Build	3- Completed	4	3	1	0	0	0	0	0	0	0	0	0	0	0	0
US-001	T-11	Prediction Endpoint – Implement /predict API endpoint that accepts stock symbol and returns Buy/Sell/Hold signal	2026-03-04 0:00:00	6-Mar-26	Gowthama R.	Coding	3- Completed	6	0	0	3	2	1	0	0	0	0	0	0	0	0	0
US-001	T-12	ML Pipeline Integration – Integrate data fetching, preprocessing, and model inference into single FastAPI pipeline	2026-03-09 0:00:00	13-Mar-26	Gowthama R.	Integration Test	3- Completed	8	0	0	0	0	0	3	3	2	0	0	0	0	0	0
SPRINT 4 BACKLOG																						
US-001	T-13	Dashboard UI Design – Create React wireframes and design layout for stock chart, signal table, and filter controls	2026-03-16 0:00:00	18-Mar-26	K. Ashok	Design	3- Completed	4	0	0	0	0	0	0	0	0	3	1	0	0	0	0
US-001	T-14	React Components – Build reusable components: stock price chart (Recharts), signal badge, and data table	2026-03-19 0:00:00	23-Mar-26	K. Ashok	Coding	3- Completed	10	0	0	0	0	0	0	0	0	0	0	4	4	2	0
US-001	T-15	Filter Controls – Add stock symbol search and date range picker; wire filters to API query params	2026-03-24 0:00:00	27-Mar-26	K. Ashok	Coding	2- In Progress	6	0	0	0	0	0	0	0	0	0	0	0	0	0	6
SPRINT 5 BACKLOG																						
US-001	T-16	Frontend–Backend Integration – Connect React to FastAPI via Axios; resolve CORS issues and handle API errors	2026-03-30 0:00:00	31-Mar-26	K. Ashok	Integration Test	2- In Progress	6	4	2	0	0	0	0	0	0	0	0	0	0	0	0
US-001	T-17	Backend Deployment – Deploy FastAPI on Render with environment variables; configure startup and keep-alive	2026-03-30 0:00:00	31-Mar-26	Karishma K.	System Test	2- In Progress	5	3	2	0	0	0	0	0	0	0	0	0	0	0	0
US-001	T-18	Frontend Deployment – Deploy React app on Vercel; configure GitHub Actions CI/CD pipeline for auto-deploy	1/Apr/26	1/Apr/26	Karishma K.	System Test	2- In Progress	5	0	0	4	1	0	0	0	0	0	0	0	0	0	0
US-001	T-19	End-to-End Testing and Documentation – Run full integration tests, fix bugs, and complete project	2/Apr/26	2/Apr/26	Gowthama R., K.	System Test	2- In Progress	4	0	0	0	0	4	0	0	0	0	0	0	0	0	0

Sprint	Day	Impediments	Action Taken
Sprint 1	Day 2	Unclear project scope and which financial APIs are most suitable.	Conducted team discussion and reviewed API documentation; selected Yahoo Finance for initial integration.
Sprint 1	Day 4	Uncertainty on architecture design – how ML model connects to backend.	Team brainstorming session held; architecture diagram created on Draw.io and approved by all members.
Sprint 2	Day 6	API rate limits encountered when fetching large historical datasets.	Implemented request throttling and caching in the data fetching script to stay within free-tier limits.
Sprint 2	Day 8	Missing values and inconsistent date formats in raw financial data.	Applied forward-fill, interpolation, and standardized date parsing using Pandas to resolve data quality issues.
Sprint 2	Day 11	Overfitting observed in initial XGBoost model during validation.	Applied cross-validation and hyperparameter tuning (learning rate, max depth) to regularize the model.
Sprint 3	Day 14	FastAPI routing conflict between /data and /predict endpoints.	Refactored API into separate router modules; each endpoint tested individually via Postman.
Sprint 4	Day 18	Recharts component causing rendering lag with large stock datasets.	Implemented data pagination and lazy loading in React; reduced initial payload size significantly.
Sprint 4	Day 20	CORS error blocking React frontend from accessing FastAPI backend.	Configured CORS middleware in FastAPI to allow the Vercel frontend origin domain.
Sprint 5	Day 23	Render free-tier backend timing out during cold starts.	Added keep-alive ping logic and optimized FastAPI startup time to reduce cold-start latency.

SL #	Sprint #	Sprint start date	Sprint end date	Team member name	Start Doing	Stop Doing	Continue Doing	Action taken
1	Sprint 1	2026-02-02 0:00:00	2026-02-13 0:00:00	Gowthama R.	Documenting architecture decisions and API integration steps in GitHub Wiki.	Assuming requirements are clear without explicit team confirmation.	Conducting daily stand-ups and using GitHub for version control.	Created shared GitHub Wiki for project documentation; all teammates reviewed architecture diagram.
2	Sprint 2	2026-02-16 0:00:00	2026-02-27 0:00:00	Leena Gavande	Logging ML model training metrics (accuracy, F1-score) after each experiment run.	Pushing untested or undocumented code directly to the main branch.	Regular pair-programming and code review sessions with Gowthama R.	Implemented structured experiment logging; enforced PR review policy before merging to main.
3	Sprint 3	2026-03-02 0:00:00	2026-03-13 0:00:00	Gowthama R.	Writing API documentation using FastAPI Swagger UI and adding inline code comments.	Hardcoding configuration values and API keys directly in source code.	Modular code design and separation of concerns across backend routers.	Moved all configurations to .env files; Swagger UI auto-documentation activated and shared with team.
4	Sprint 4	2026-03-16 0:00:00	2026-03-27 0:00:00	K. Ashok	Adding loading spinners and error boundaries to all React API-dependent components.	Mixing API call logic directly inside UI components instead of separating into service files.	Component-based React architecture with reusable UI elements across the dashboard.	Refactored API calls into a dedicated services layer; loading and error states added to all components.
5	Sprint 5	2026-03-30 0:00:00	2026-04-02 0:00:00	Gowthama R, K. Ashok, Karishma K, Leena Gavande	Running end-to-end integration tests before each deployment to catch regressions early.	Waiting until the final sprint day to perform integration and deployment testing.	Agile sprint retrospectives, open team communication, and collaborative debugging sessions.	All integration tests passed; platform deployed successfully on Render and Vercel on schedule.